02DataSplit

February 6, 2026

[1]:
```
# import packages
import pandas as pd
import numpy as np

# read in unrefined dataset
tourney_games = pd.read_csv("tourneyGames.csv")
```

[2]:
```
# drop unnecessary columns
cols_to_drop = list(range(12,13)) + list(range(101, 157)) + list(range(159,␣
 ↪172)) + list(range(260, 316)) + list(range(318, 330))
tourney_games = tourney_games.drop(tourney_games.columns[cols_to_drop], axis=1)
```

[3]:
```
# create target (1 if better seed won, 0 if worse seed won)
def lower_seed_won(row):
    if row["winnerTeamId"] == row["team1Id"]:
        winning_seed = row["team1_Seed"]
        losing_seed = row["team2_Seed"]
    else:
        winning_seed = row["team2_Seed"]
        losing_seed = row["team1_Seed"]

    return int(winning_seed < losing_seed)

tourney_games["y"] = tourney_games.apply(lower_seed_won, axis=1)

# identify team stat columns
team1_cols = [c for c in tourney_games.columns if c.startswith("team1_")]
team2_cols = [c for c in tourney_games.columns if c.startswith("team2_")]

# find shared stats between team1 and team2
stats = sorted(
    set(c.replace("team1_", "") for c in team1_cols)
    & set(c.replace("team2_", "") for c in team2_cols)
)

# create difference stats (team1 - team2)
for stat in stats:
```

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    tourney_games[f"{stat}_diff"] = tourney_games[f"team1_{stat}"] -␣
 ↪tourney_games[f"team2_{stat}"]

# keep identifier data
base_cols = [
    "Season", "DayNum",
    "WTeamID", "LTeamID",
    "WScore", "LScore",
    "team1Id", "team2Id",
    "y", "winnerTeamId",
    "team1Name", "team2Name",
    "team1_Seed", "team2_Seed"
]

# keep difference columns
diff_cols = [f"{stat}_diff" for stat in stats]

# final dataset creation
final_df = tourney_games[base_cols + diff_cols]

final_df
```

[3]:

	Season	DayNum	WTeamID	LTeamID	WScore	LScore	team1Id	team2Id	y	\
0	2003	134	1421	1411	92	84	1411	1421	0	
1	2003	136	1112	1436	80	51	1112	1436	1	
2	2003	136	1113	1272	84	71	1113	1272	0	
3	2003	136	1141	1166	79	73	1141	1166	0	
4	2003	136	1143	1301	76	74	1143	1301	1	
…	…	…	…	…	…	…	…	…	..	
1348	2024	146	1301	1181	76	64	1181	1301	0	
1349	2024	146	1345	1397	72	66	1345	1397	1	
1350	2024	152	1163	1104	86	72	1104	1163	1	
1351	2024	152	1345	1301	63	50	1301	1345	1	
1352	2024	154	1163	1345	75	60	1163	1345	0	

	winnerTeamId	…	Raw Defensive Efficiency	Rank_diff	\
0	1421	…		-165	
1	1112	…		-34	
2	1113	…		123	
3	1141	…		159	
4	1143	…		-39	
…	…	…	…		
1348	1301	…		-106	
1349	1345	…		45	
1350	1163	…		261	
1351	1345	…		85	
1352	1163	…		-43	

```
     Raw Offensive Efficiency_diff  Raw Offensive Efficiency Rank_diff  \
0                              1.6                                -43
1                              9.0                               -133
2                              3.2                                -42
3                             -5.4                                 34
4                             -1.7                                 19
…                              …                                  …
1348                           8.4                                -90
1349                           7.1                                -40
1350                          -1.6                                  1
1351                         -10.3                                 97
1352                           3.7                                 -3

      Raw Tempo_diff  Raw Tempo Rank_diff  Seed_diff  StlRate_diff  \
0                1.1                  -36          0     -0.009500
1               10.0                 -263        -15      0.006100
2               -0.7                   25          3     -0.026100
3                3.3                 -100          5     -0.020000
4                2.2                  -96         -1     -0.020600
…                …                     …          …             …
1348            -1.4                   95         -7     -0.007230
1349            -1.9                  114         -1     -0.031880
1350             7.9                 -296          3     -0.000093
1351             0.2                  -22         10      0.025653
1352            -2.3                  115          0      0.014265

      TOPct_diff  Tempo_diff  eFGPct_diff
0      -1.005500      1.1023     1.696500
1      -2.422900      9.9893     2.742900
2       0.126500     -0.6498     1.957200
3       5.750300      3.3585    -0.099900
4      -1.314400      2.1439    -1.044500
…            …           …           …
1348    0.653412     -1.4484     3.893735
1349    1.966710     -1.9514     4.685253
1350    1.418475      7.8314    -0.594558
1351   -2.647674      0.2654    -5.084930
1352   -1.943160     -2.2433     1.097619

[1353 rows x 104 columns]
```

[4]:
```
# export dataset to csv
final_df.to_csv("matchupDiff.csv")
```

[5]:
```
# split into training and testing
train_df = final_df[(final_df["Season"] >= 2003) & (final_df["Season"] <= 2019)]
```

```
test_df  = final_df[(final_df["Season"] >= 2021) & (final_df["Season"] <= 2024)]

# separate features and target
X_train = train_df.drop(columns=["y"])
y_train = train_df["y"]

X_test = test_df.drop(columns=["y"])
y_test = test_df["y"]
```

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